
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection constitutes a foundational prerequisite for the
2 robustness and reliability of machine learning systems deployed within real-world
3 applications. Although a recent line of inquiry has interrogated the exploitation
4 of unlabeled data and thereby intimated its promise for augmenting OOD detec-
5 tion capabilities, the effective harnessing of unlabeled in-the-wild data remains a
6 formidable undertaking, owing to the heterogeneous admixture of in-distribution
7 (InD) and OOD samples that such data comprise. The absence of a cleanly delin-
8 eated set of OOD exemplars considerably complicates the endeavour of training an
9 optimal OOD classifier. In the present work, we introduce Medix, a novel frame-
10 work devised to isolate prospective outliers from unlabeled data through recourse
11 to the median operation—an operator we adopt precisely because, by furnishing a
12 stable estimate of central tendency that is intrinsically resilient to noise and extreme
13 values, it lends itself naturally to an OOD detection mechanism. Leveraging these
14 isolated outliers in concert with the labeled InD data, we subsequently train a robust
15 OOD classifier. From a theoretical standpoint, we derive error bounds that establish
16 the low error rate attained by Medix, and empirical findings further corroborate
17 these assertions: Medix surpasses extant methods uniformly across open-world
18 settings, thereby vindicating the validity of our theoretical insights.

19 1 Introduction

20 The deployment of machine learning models within real-world applications routinely subjects them
21 to challenges of safety and reliability, a predicament attributable in no small part to the presence of
22 out-of-distribution (OOD) data. Such OOD samples, which originate from categories unbeknownst
23 to the model, ought not to elicit any prediction; yet neural networks, being inherently susceptible in
24 this regard, characteristically lack the mechanisms requisite for detecting and duly handling OOD
25 inputs in practice (Nguyen et al., 2015).

26 The identification of OOD samples at inference time, though indispensable, is intrinsically arduous,
27 since models are never exposed to unknown distributions over the course of training and, in conse-
28 quence, cannot reliably discriminate OOD from in-distribution (InD) data. In an effort to surmount
29 this difficulty, a recent body of work (Katz-Samuels et al., 2022a; Du et al., 2024a) has explored
30 the exploitation of supplementary “in-the-wild” data as a means of ameliorating OOD detection. In
31 particular, Katz-Samuels et al. (2022a) advanced a method that enlists unlabeled wild data for the
32 regularization of model training while retaining its focus on the classification of labeled InD data.
33 The salient advantage of such unlabeled wild data resides in its ready availability—it is trivially
34 collectible once a model has been deployed within its operating environment—an arrangement that
35 affords the model a more faithful apprehension of the true distribution of OOD data encountered at
36 test time and thereby engenders more robust OOD detection.

37 The exploitation of unlabeled wild data, however, is far from unencumbered, precisely on account
 38 of the intricate commingling of InD and OOD data that it entails. The want of a distinct and
 39 uncontaminated set of OOD samples confounds the development of robust OOD detection methods,
 40 all the more so given that the OOD detector is confronted solely with data drawn from this mixed
 41 distribution and remains oblivious to whether any given sample issues from the InD or the OOD
 42 category. As matters presently stand, the problem persists in a comparatively underexplored state,
 43 leaving substantial latitude for further advancement; moreover, few studies furnish a formal theoretical
 44 foundation, and to the best of our knowledge Du et al. (2024a) constitutes the sole work to supply
 45 such a foundation for the “in-the-wild” setting.

46 Concurrently, recent investigations have evidenced the efficacy of median-based approaches in the
 47 context of data pruning (Acharya et al., 2024). Impelled by these developments, the present paper
 48 undertakes to address the following question:

How can median-based methods leverage unlabeled wild data to facilitate OOD
 detection with theoretical guarantees?

49 In pursuit of an affirmative response to this question, we introduce a novel median-centric perspective
 50 on OOD detection. Concretely, we formulate a median-based optimization framework and develop an
 51 accompanying algorithm, Medix, which discerns OOD samples from wild data with a demonstrably
 52 low error rate. We furnish a theoretical foundation that guarantees minimal error and, through
 53 experiments that attest to the robustness and efficiency of our algorithm, corroborate that foundation
 54 empirically; ours is among the few studies to establish such a foundation for the unlabeled “in-the-
 55 wild” setting. We demonstrate that median-based filtering constitutes a robust instrument for outlier
 56 detection within unlabeled mixtures by bounding the fraction of InD samples erroneously flagged
 57 as outliers—a fraction that is, in turn, governed by two distinct effects: the *contamination effect*,
 58 which quantifies the influence exerted by OOD points and remains tractable provided that the OOD
 59 proportion falls below 50%, and the *concentration effect*, which exploits the sub-Gaussian character
 60 of InD gradients so as to bound the deviations of InD points from their mean gradient. Acting in
 61 concert, these effects ensure that, with high probability, the number of misclassified InD points
 62 remains small, thereby evidencing the efficacy of the method even under worst-case conditions.

63 We benchmark our approach against two categories of methods: those trained exclusively on InD data,
 64 on the one hand, and those trained on InD data in conjunction with an auxiliary unlabeled dataset, on
 65 the other. On CIFAR-100 (Krizhevsky et al., 2009), Medix exhibits a pronounced improvement over
 66 the formidable baseline KNN+ (Sun et al., 2022), eclipsing it by an average of 40.98% with respect to
 67 FPR95. In contradistinction to approaches such as Outlier Exposure (Hendrycks et al., 2019)—which
 68 are predicated upon a clean, auxiliary unlabeled dataset and thereby invoke the stringent distributional
 69 assumption that the auxiliary data are entirely separable from the InD data—Medix attains superior
 70 results in the absence of any such assumption, thereby conferring appreciably greater flexibility.
 71 Relative to WOODS (Katz-Samuels et al., 2022a), Medix curtails the average FPR95 by 1.32% on
 72 CIFAR-100 and by 2.60% on CIFAR-10.

73 Our key contributions are as follows:

- 74 **C1)** We propose Medix, a median-centric greedy procedure that first filters outliers from the
 75 unlabeled wild data and subsequently trains an OOD detector upon the conjunction of
 76 the identified outliers and the InD samples; the filtering stage constitutes our principal
 77 contribution.
- 78 **C2)** We establish theoretical guarantees pertaining to the robustness of median-based filtering
 79 in the identification of both inliers and outliers within unlabeled mixtures. Specifically, we
 80 prove that the misclassification rates—both for InD samples spuriously flagged as outliers
 81 and for OOD samples inadvertently retained as inliers—are tightly circumscribed by two
 82 effects: the contamination effect, which remains bounded so long as the proportion of OOD
 83 samples does not exceed 50%, and the concentration effect, which underwrites the stability
 84 of the gradient behaviour exhibited by InD samples.
- 85 **C3)** We undertake an extensive evaluation of Medix spanning eleven InD-OOD pairs, juxtaposing
 86 its performance against 20 competitive baselines. Our results demonstrate that Medix
 87 surpasses the entirety of these baselines, attaining superior performance across the board; we
 88 further show, by way of experiment, that the outlier extraction performed by Medix achieves

a low error rate (a mere 12.5%, for instance; see Figure 2), a finding that corroborates our theoretical analysis.

2 Preliminaries and Problem Setup

In this section we first furnish an overview of the OOD detection problem and thereafter formally delineate the data setup, the model architecture, the loss functions, and the overarching learning objective.

Labeled In-Distribution Data. Consider the input space $\mathcal{X}$ together with the label space $\mathcal{Y} = \{1, \dots, K\}$, which jointly circumscribe the structure of the InD data. A labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ is engendered by drawing n pairs independently and identically distributed (i.i.d.) from $\mathbb{P}_{XY}$, an unknown joint probability distribution over $\mathcal{X} \times \mathcal{Y}$. The marginal distribution induced by $\mathbb{P}_{XY}$ on $\mathcal{X}$, which we denote $\mathbb{P}_{\text{in}}$, represents the underlying distribution of InD inputs, and it is $\mathcal{S}_{\text{in}}$ that we employ to train an InD model.

Out-of-distribution detection. We address a practical scenario in which the model, though trained upon labeled InD data, is subsequently deployed within environments liable to harbour OOD inputs drawn from classes unrepresented in the training data—that is, associated with some label $y \notin \mathcal{Y}$ —and is expected to abstain from rendering predictions for such inputs. At inference time, the paramount objective is to adjudicate whether a given input belongs to the InD distribution or instead arises from an OOD source.

Unlabeled wild data. Among the foremost impediments confronting OOD detection is the scarcity of labeled OOD samples. Because the potential sample space of OOD data may be exceedingly vast, the collection of labeled exemplars proves at once costly and impractical. To circumvent this obstacle, we incorporate into our learning framework unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, so as to more faithfully emulate real-world scenarios, following the proposal of Katz-Samuels et al. (2022a). Such wild data constitute an admixture of InD and OOD samples and may be readily amassed during the deployment phase of a model pre-trained on $\mathcal{S}_{\text{in}}$. In keeping with Du et al. (2024a); Katz-Samuels et al. (2022a), we adopt the Huber contamination model (Huber, 1964) to characterize the marginal distribution of the wild data

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ captures the OOD distribution over $\mathcal{X}$. We remark that the degenerate case $\pi = 0$, corresponding to the outright absence of OOD samples, renders the problem trivial.

Models and Loss Functions. Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ denote the InD classifier parameterized by $\phi \in \Phi$, wherein Φ designates the attendant parameter space; the output of f_ϕ constitutes a soft probability distribution over the $K = |\mathcal{Y}|$ InD classes, and the loss function governing the labeled InD data is specified as $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For the purpose of OOD detection, we introduce a separate classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$, parameterized by $\theta \in \Theta$ with Θ its parameter space, whose associated binary loss function we denote $\ell_b(g_\theta(x), y_b)$, where $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$; here $y_+ > 0$ signifies the InD class, whereas $y_- < 0$ corresponds to the OOD class.

Learning objective. Our learning framework is contrived so as to train the OOD detector g_θ and the multi-class classifier f_ϕ concurrently, drawing upon both the InD data $\mathbb{P}_{\text{in}}$ and the wild data $\mathbb{P}_{\text{wild}}$. At test time, we appraise performance by recourse to the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{XY}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

wherein $\mathbb{I}\{\cdot\}$ denotes the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ designates the OOD test-data distribution.

3 Method: Median-Centric Framework for OOD Detection

In this section we expound a novel learning paradigm, termed **Medix**, conceived for OOD detection through the harnessing of unlabeled wild data. Transcending the limitations that beset conventional

approaches predicated exclusively on InD data, our framework is especially well-suited to applications in open-world environments, wherein models are frequently confronted with hitherto unseen inputs. The Medix framework comprises two integral stages: **1) Outlier Extraction**, a filtering process that isolates candidate OOD samples from the unlabeled wild data (elaborated in Section 3.1), and **2) Detector Training**, wherein a binary OOD detector is trained upon both the InD data and the outlier candidates identified at the preceding step (elaborated in Section 3.2); for this second stage we adhere to the protocol advanced by Du et al. (2024a). As we shall demonstrate in the sections that ensue, this two-step methodology not only facilitates the effective extraction of OOD data from the unlabelled wild data but likewise establishes a robust foundation for the deployment of machine learning models within dynamic, open-world scenarios.

3.1 Extracting Candidate Outliers from the Wild Data

In order to isolate prospective outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, our framework marshals an optimization-based approach that capitalizes on the gradients of the model parameters—gradients that are themselves derived from a classification model, f_ϕ , trained solely upon the InD dataset $\mathcal{S}_{\text{in}}$. The detailed methodology underlying this process is formally delineated below.

Reference gradient estimation from InD data. The inaugural step of our proposed framework consists in estimating a reference gradient by means of the InD dataset $\mathcal{S}_{\text{in}}$, an estimation effected by training a classifier f_ϕ on $\mathcal{S}_{\text{in}}$ through empirical risk minimization (ERM) as follows

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \text{ where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. Once the classifier has been trained, we compute the mean gradient $\bar{\nabla}_{\text{in}}$, defined as the average, taken over the InD data, of the gradients of the loss function with respect to the model parameters:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

Within our approach, $\bar{\nabla}_{\text{in}}$ serves as the reference gradient, thereby enabling the quantification of the deviations exhibited by other data points relative to this reference.

Motivation. We posit the hypothesis that augmenting the number of OOD samples within the wild dataset $\mathcal{S}_{\text{wild}}$ will engender a commensurately greater deviation from the average InD gradient $\bar{\nabla}_{\text{in}}$. To interrogate this hypothesis, we devise a preliminary experiment that employs CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. Specifically, $\mathcal{S}_{\text{wild}}$ comprises 10,000 samples drawn from CIFAR-10, so constituted as to be disjoint from the training set used to train the model $\phi_{\mathcal{S}_{\text{in}}}$ —the very model from which we compute $\bar{\nabla}_{\text{in}}$. Incrementally introducing SVHN OOD samples into $\mathcal{S}_{\text{wild}}$, we track the evolution of the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the gradients of the wild dataset, as given by:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

Our results, portrayed in Figure 1, disclose a pronounced and monotonic ascent in the L_2 -norm deviation, thereby lending support to our hypothesis. This observation furnishes a pivotal motivation for the method we introduce in the ensuing section. Notably, the stopping criterion of our algorithm is itself derived from this monotonically increasing behaviour: we terminate the procedure once the L_2 -norm deviation between consecutive iterations descends below a threshold ϵ —a mechanism we shall expound in detail in what follows.

Filtering potential outliers from unlabeled wild data. Motivated by the results reported in Figure 1, we formulate the following optimization problem for the purpose of identifying the outlier subset $\mathcal{S}_{\text{out}}^*$ within $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function, while $\hat{y}_{\tilde{\mathbf{x}}_i}$ designates the predicted label for a wild sample $\tilde{\mathbf{x}}_i$; for the sake of notational economy, we define $G_{\mathcal{S}} = \{ \nabla \ell(\tilde{\mathbf{x}}_i) \}_{i \in \mathcal{S}}$.

176 The optimization problem set forth above seeks to
 177 identify a subset $\mathcal{S}$ of $\mathcal{S}_{\text{wild}}$ that minimizes the distance
 178 between the EWM of the gradients and the
 179 average gradient $\bar{\nabla}_{\text{in}}$. As Figure 1 intimates, such
 180 a subset, denoted $\mathcal{S}_{\text{in}}^*$, may plausibly represent the
 181 InD data residing in $\mathcal{S}_{\text{wild}}$, in which event its complement
 182 $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ may correspondingly
 183 capture the OOD data.

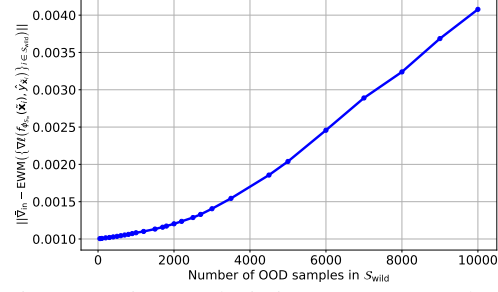


Figure 1: Distance deviation as OOD samples are progressively introduced into $\mathcal{S}_{\text{wild}}$.

184 Solving the optimization problem in (4) can prove
 185 computationally prohibitive, particularly as the cardinality of $\mathcal{S}_{\text{wild}}$ grows. To surmount this obstacle,
 186 we advance a greedy approximation founded upon a
 187 leave-one-out strategy, as delineated in Algorithm 1. The algorithm instantiates an iterative procedure
 188 for outlier detection from a wild dataset, one that capitalizes on the deviations in gradient information
 189 relative to the InD dataset.

191 The algorithm commences, at each iteration, by computing the EWM of the wild-data gradients,
 192 which thereupon serves as a reference against which the average gradient of the InD data $\bar{\nabla}_{\text{in}}$ is
 193 compared. We denote by d_t the L_2 distance between the average InD gradient $\bar{\nabla}_{\text{in}}$ and the EWM of
 194 the gradients of the data remaining in the wild set $\mathcal{S}$; that is, $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$. The algorithm
 195 then iteratively singles out, for designation as OOD, those samples in $\mathcal{S}$ whose removal precipitates
 196 the most substantial diminution in the L_2 distance relative to d_t . Concretely, given the samples $\mathcal{S}$ that
 197 remain, we excise each sample $i \in \mathcal{S}$, compute the resultant $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ together with its distance
 198 $\|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ to $\bar{\nabla}_{\text{in}}$, evaluate the attendant reduction $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$,
 199 and identify as OOD the k samples occasioning the largest such reduction. This procedure recurs
 200 until either no appreciable reduction in δ_i subsists or a maximum number of iterations has been
 201 attained. The convergence criterion hinges upon the change in the L_2 distance across two successive
 202 iterations, which must fall below a predefined threshold ϵ ; this criterion is itself inspired by the
 203 monotonically increasing trend observed in our preliminary experiment (Figure 1), inasmuch as, once
 204 a substantial contingent of OOD samples has been removed, the L_2 distance gradually subsides to a
 205 small value—signalling the juncture at which the algorithm ought to halt so as to avert the erroneous
 206 designation of InD samples as outliers and thereby forestall any degradation in performance.

Algorithm 1 Iterative Outlier Detection via Medix

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i}})$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$
 (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L_2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}, t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\hat{\mathcal{S}}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

207 3.2 Training the OOD Detector with Candidate Outliers

208 Having identified the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ from the wild data, we proceed to train the OOD
 209 detector g_{θ} , which is engineered to maximize the distinction between InD data and candidate outlier
 210 data in accordance with the protocol of Du et al. (2024a). The objective function explicitly enforces
 211 separability at the decision boundary (thresholded at 0), assigning positive outputs to labeled InD
 212 samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs to candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. Specifically, the loss function

213 is defined as:

$$\begin{aligned}
\mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_\theta) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta), \\
\mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_\theta(\mathbf{x}) \leq 0\}, \\
\mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_\theta(\tilde{\mathbf{x}}) > 0\}.
\end{aligned} \tag{5}$$

214 To contend with the challenge posed by the non-differentiable character of the 0/1 loss, we enlist
215 a binary loss grounded in a differentiable sigmoid function as a smooth surrogate, which ensures
216 tractable optimization while preserving fidelity to the original objective. The learning framework
217 further incorporates the InD risk defined in Equation 2, thereby safeguarding predictive accuracy on
218 InD samples. This unified optimization scheme materially fortifies the generalization performance of
219 g_θ and, as our results corroborate, empowers the robust detection of OOD samples drawn from $\mathbb{P}_{\text{out}}$.

220 4 Theoretical Analysis

221 We now present the theoretical guarantees attending the filtering stage of Medix. The theorems that
222 follow supply provable upper bounds on the misclassification rates for both InD and OOD points and,
223 taken together, evince the two-sided robustness of our EWM filtering. Detailed proofs are deferred to
224 Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

225 Theorem 4.1 discloses that the inlier misclassification rate is governed by two salient effects: a
226 *concentration term*, which vanishes as the sample size increases and which encapsulates the sta-
227 tistical fluctuations of the InD gradients, and a *contamination term*, which quantifies the influence
228 exerted by OOD points upon the EWM. Notably, the bound remains under control so long as the
229 contamination ratio satisfies $\pi < 0.5$, a fact that underscores the resilience of median-based filtering
230 to a moderate OOD presence. We now turn to the complementary question—*how many true outliers*
231 *elude detection and are mistakenly retained?*—which the ensuing theorem resolves by bounding the
232 OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

233 Theorem 4.2 complements the foregoing result by bounding the fraction of OOD points mistakenly
234 retained as InD. In conjunction, Theorems 4.1 and 4.2 furnish a two-sided guarantee for the robustness
235 of the Medix filtering method, the inlier and outlier misclassification rates being governed by an
236 equilibrium among the following three effects:

237 **Contamination Effect.** As concerns Theorem 4.1, even when the wild dataset harbours a considerable
238 fraction π of OOD points, the median-based criterion retains its stability provided that $\pi < 0.5$.

This stability is reflected in the $\pi/[2(1 - \pi)]$ term of the bound, which grows with π and thereby encodes the risk that OOD gradients may skew the EWM. Conversely, Theorem 4.2 incorporates a symmetric penalty $\frac{1-\pi}{2\pi}$, which quantifies the difficulty of isolating OOD points whenever they are underrepresented.

Concentration Effect. Both bounds capitalize upon the sub-Gaussian nature of the gradient coordinates. In the case of InD data, this property ensures that the preponderance of gradients concentrate in the vicinity of $\bar{\nabla}_{\text{in}}$, thereby keeping the median stable even in the finite-sample regime; in the case of OOD points, concentration about μ_{out} permits us to quantify the risk that a misaligned OOD sample may slip past the filter. These concentration effects decay at the rate $1/\sqrt{m}$, where m denotes the number of samples drawn from each class.

Separation Effect. Peculiar to Theorem 4.2, this effect quantifies the extent to which the OOD mean gradient must lie apart from the InD mean gradient if OOD samples are to be reliably rejected. The exponential term $\exp(-(\Delta - \varepsilon)^2/2\sigma_{\text{out}}^2)$ encapsulates precisely this trade-off: the more markedly the distributions are separated, the less probable it becomes that OOD gradients will deceive the filter.

Theorems 4.1 and 4.2 jointly establish that, with high probability, median filtering effects a robust separation of InD and OOD samples within unlabeled data mixtures. Whereas the fraction of InD samples misclassified as outliers is circumscribed by the contamination and concentration effects, the fraction of OOD points erroneously retained is governed by an exponential separation term, a concentration bound, and a reverse contamination effect. Collectively, these results furnish a rigorous theoretical assurance that Medix minimizes errors of both kinds under mild assumptions.

Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates. The supposition that the gradients of InD samples are sub-Gaussian in each coordinate is indispensable both for the derivation of concentration bounds and for the assurance of robustness in statistical estimation. Additional empirical evidence, adduced in Figure 4a, substantiates this supposition: therein we observe that the histogram of gradient values for InD samples is bell-shaped and concentrated about the mean, an observation indicating that the gradients possess light tails—the defining hallmark of sub-Gaussian random variables, whose tails decay exponentially and for which, accordingly, extreme gradient values are seldom encountered. Furthermore, Figure 4b juxtaposes the empirical quantiles of the gradient distribution against the theoretical quantiles of a Gaussian distribution by way of a Q-Q plot (Wilk and Gnanadesikan, 1968); the close adherence of the points to the 45-degree reference line demonstrates that the empirical distribution of the gradients indeed approximates a Gaussian distribution closely and, inasmuch as Gaussian random variables constitute a subclass of the sub-Gaussian variables, lends further credence to the sub-Gaussian assumption. *We additionally furnish a looser rendition of these bounds that dispenses with the sub-Gaussian assumption and holds under merely bounded second moments; this rendition is presented in Theorem C.3 (Appendix C.3). Although the rates therein degrade, the core robustness guarantee of Medix persists within this broader setting.*

5 Experiments

This section undertakes to demonstrate the efficacy of Medix across a variety of InD-OOD dataset pairs, benchmarking it against 20 widely adopted baselines. The entirety of our experiments is conducted on hardware furnished with NVIDIA A100-SXM4-80GB GPUs, and the code requisite for the reproduction of our results will be released upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. We adopt the identical experimental protocol employed by Katz-Samuels et al. (2022a), who inaugurated the problem of learning OOD detectors with wild data, so as to enable an equitable comparison with prior work. Concretely, we designate CIFAR-10 and CIFAR-100 as the InD datasets ($\mathbb{P}_{\text{in}}$), while for OOD testing we select a suite of natural-image datasets—namely PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015)—to serve as the OOD datasets ($\mathbb{P}_{\text{out}}$). To simulate wild data ($\mathbb{P}_{\text{wild}}$), we amalgamate a subset of the InD data ($\mathbb{P}_{\text{in}}$) with the OOD data ($\mathbb{P}_{\text{out}}$) under a default mixing parameter $\pi = 0.5$. By way of

illustration, when PLACES365 serves as the OOD test set, we constitute a wild mixture by combining CIFAR with PLACES365 and thereafter test on PLACES365 as the OOD set—a procedure reiterated across the full complement of OOD datasets and baselines. The InD CIFAR dataset is partitioned into two halves: the first 25,000 images are apportioned to the training of $\phi_{\mathcal{S}_{\text{in}}}$, whereas the remainder are reserved for the generation of the wild mixture $\mathcal{S}_{\text{wild}}$. For the computation of gradients we avail ourselves of the penultimate-layer weights, which have been shown to be especially informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We conduct our evaluation by recourse to three standard metrics: (1) the False Positive Rate (FPR95 $\downarrow$) of OOD samples at the operating point where the True Positive Rate of InD samples equals 95%, (2) the Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and (3) the InD Classification Accuracy (InD Acc $\uparrow$).

Baselines. Our comparison spans a heterogeneous assortment of competitive baselines, stratified according to whether they are trained upon InD data alone or upon both InD and wild data. For the methods trained exclusively on InD data ($\mathbb{P}_{\text{in}}$), we set Medix against a variety of well-established OOD detection methods—among them Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022)—all of which employ a model trained under the softmax cross-entropy loss; for the sake of a more comprehensive comparison, we further contrast Medix with methods grounded in a contrastive loss, such as CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). As regards the methods that draw upon both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), both of which regularize training by promoting lower confidence or higher energy on outlier data. We likewise incorporate a comparison with WOODS (Katz-Samuels et al., 2022a), which introduced the notion of wild unlabeled data and exploits it for OOD detection through a constrained optimization approach. Finally, so as to render the evaluation more thorough, we include the more recent baselines CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024).

5.2 Experimental Setup

In consonance with WOODS (Katz-Samuels et al., 2022a), we adopt for the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ a Wide ResNet architecture (Zagoruyko, 2016) of 40 layers with a width factor of 2, trained by stochastic gradient descent with a momentum of 0.9, a weight decay of 0.0005, and an initial learning rate of 0.1; training proceeds for 100 epochs under cosine learning-rate decay, with a batch size of 128 and a dropout rate of 0.3. The hyperparameters ϵ and k governing the proposed method Medix are chosen from the sets $\{5\text{e-}5, 5\text{e-}4, 5\text{e-}3, 5\text{e-}2\}$ and $\{4\text{k}, 7\text{k}, 10\text{k}, 20\text{k}\}$, respectively, with due regard for the dataset sizes and with the aim of maximizing OOD performance. As for the OOD classifier g_θ , we initialize it with the pre-trained InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ and append a linear layer that performs binary classification upon the penultimate-layer features; the learning rate of this classifier is fixed at 0.001, and fine-tuning is carried out over 100 epochs in accordance with Equation (5). We conjoin the binary classification loss with the InD classification loss, assigning a weight of 10 to the binary classification component, while all remaining training parameters coincide with those employed in the training of $\phi_{\mathcal{S}_{\text{in}}}$.

5.3 Results

We report our principal results on CIFAR-100 in Table 2, wherein Medix conspicuously outperforms the whole of the OOD detection baselines. These results bring the following salient observations into relief: (1) methods trained upon both InD and wild data substantially eclipse those trained exclusively on InD data, Medix curtailing the FPR95 by 52.31% on PLACES365 and by 38.24% on TEXTURES relative to KNN+ and thereby attesting to the efficacy of incorporating in-the-wild data for the regularization of the model; (2) Medix moreover surpasses even the competitive methods that exploit wild data ($\mathbb{P}_{\text{wild}}$): on CIFAR-100 it attains an average FPR95 of 5.42%, an improvement of 1.32% over WOODS, while sustaining a competitive InD accuracy of 73.33%—a marginal shortfall attributable to the circumstance that our method is trained on 25,000 labeled InD samples, whereas the baselines, which forgo wild data, avail themselves of the full CIFAR-100 training set of 50,000 samples. We further present the CIFAR-10 results in Table 1, where Medix likewise surpasses every

Table 1: Comparison of OOD detection performance between Medix and the baselines with CIFAR-10 as the InD data. Performance is averaged over five runs, and the best results in each column are rendered in **bold**.

Methods	OOD Datasets												InD Acc $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_m$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	89.57	81.44	90.44	94.38	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_m$ and $\mathbb{P}_{wild}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: Comparison of OOD detection performance between Medix and the baselines with CIFAR-100 as the InD data. Performance is averaged over five runs, and the best results in each column are rendered in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_m$ only													
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using $\mathbb{P}_m$ and $\mathbb{P}_{wild}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

baseline method, exceeding WOODS by 7.21% on PLACES365 and by 5.25% on TEXTURES with respect to FPR95 and thereby evidencing its efficacy in the detection of OOD samples.

A representative visual example of Medix. We proceed to interrogate the performance of Algorithm 1 (Outlier Extraction) in extracting OOD samples from the wild data $\mathcal{S}_{wild}$. So as to render this behaviour visualizable, we contrive an experiment upon 2-dimensional synthetic data, a simulation deliberately kept simple in the interest of expository clarity. We generate the InD data by sampling from three multivariate Gaussian distributions corresponding to three classes, whose mean vectors are set respectively to $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$, with the covariance matrix of all three classes fixed at $0.25 \cdot I$; for the OOD data, we employ a multivariate Gaussian distribution $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. As Figure 2 reveals, our method successfully identifies the outliers, the preponderance of which coincide with the ground truth: Medix flags 87.5% of the genuine OOD samples as outliers, a result that underscores its robustness in outlier extraction.

Additional studies and insights. Owing to constraints of space, we relegate supplementary experiments and insights to Appendix A, which encompass (1) ablation studies concerning hyperparameter selection together with a sensitivity analysis (Appendix A.2), which attest to the pronounced robustness of Medix to its hyperparameters; (2) a comparison between the EWM and the geometric median (Appendix A.1), which establishes that the EWM is the more sensitive to distributional shifts and hence the more effective and reliable choice for the filtering employed in our method; (3) a comprehensive comparison with methods predicated upon competitive contrastive learning objectives (Appendix A.3), which demonstrates that Medix outstrips even these competitive baselines by a substantial margin; (4) an evaluation on large-scale data under the demanding unseen OOD setting, wherein $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4), which likewise demonstrates that Medix outperforms the baselines by a considerable margin; (5) an assessment of the computational and memory efficiency of Medix (Appendix A.6); (6) an examination of the influence of pseudo-label quality (Appendix A.5),

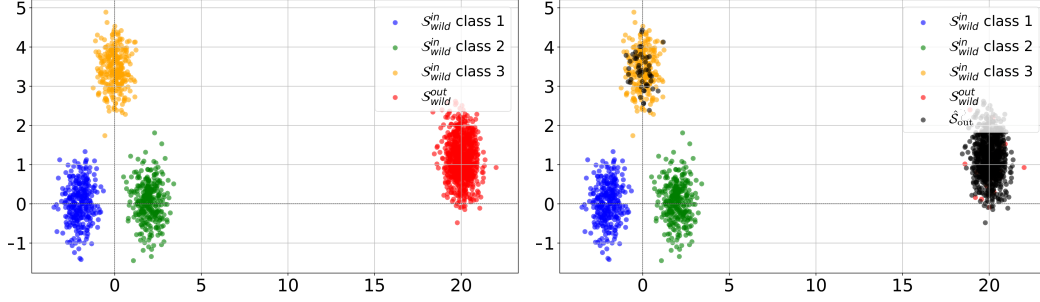


Figure 2: An illustrative application of Medix to unlabeled wild data. (a) The configuration of the InD data $S_{\text{wild}}^{\text{in}}$ and the OOD data $S_{\text{wild}}^{\text{out}}$ within the wild set, the inliers being sampled from three multivariate Gaussian distributions. (b) The outliers $\hat{S}_{\text{out}}$ filtered by Medix (depicted in black), for which the error rate—the fraction of $\hat{S}_{\text{out}}$ that in fact comprises InD data $S_{\text{wild}}^{\text{in}}$ —amounts to a mere 12.5%.

which reveals that our method is resilient to noisy or low-confidence labels; and (4) a comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

Recent years have witnessed a burgeoning interest in OOD detection (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djuricic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One approach to the detection of OOD data enlists scoring functions to appraise the data distribution, a category that subsumes distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Sehwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

A further approach to OOD detection has recourse to regularization techniques applied over the course of the training phase (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). Such techniques may, for instance, be brought to bear upon the model so as either to attenuate its confidence (Lee et al., 2017; Hendrycks et al., 2019) or to elevate its energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on the OOD data; the preponderance of these regularization methods, however, presuppose the availability of an auxiliary OOD dataset.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) have relaxed the assumption of employing labeled data alone by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), albeit without proposing any clear mechanism for outlier detection. In contradistinction, Du et al. (2024a,b) introduced an explicit outlier filtering method; yet their thresholding technique diverges from ours in a fundamental respect, inasmuch as we deploy a novel median-centric approach to the detection of outliers. Moreover, Katz-Samuels et al. (2022a); Du et al. (2024a) operate under the presumption of batch-level mixing, whereby each batch exhibits a fixed ratio of InD to OOD samples—a presumption that founders in the presence of large outsourced datasets, wherein the data are commingled at random across the dataset rather than according to any such structure; our method obviates this difficulty by accommodating dataset-level mixing without reliance upon batch-level structure. A substantial body of work additionally draws upon positive-unlabeled learning, which trains classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al.,

2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022). A cardinal distinction from our approach, nonetheless, is that these methods are preoccupied solely with discriminating $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ and do not concurrently train an OOD classifier; we, by contrast, propose a median-centric method for identifying outliers in unlabeled data that is furnished with provably low error rates.

7 Conclusions

In the present work we have introduced Medix, a novel median-centric framework for OOD detection that draws upon unlabeled in-the-wild data. By capitalizing on the intrinsic robustness of the median operation, Medix filters outliers from mixed unlabeled data with efficacy and thereby renders possible the training of a reliable OOD detector. Our theoretical analysis established provable bounds on the inlier misclassification rate, demonstrating that Medix preserves its robustness even under substantial OOD contamination (up to 50%), with the errors held in check by sub-Gaussian concentration and contamination effects; we further supplied complementary theoretical limits on the rate of OOD misclassification, so as to accurately isolate OOD samples under conditions of clear separation. Empirical validation spanning a diversity of benchmarks laid bare the performance ascendancy of Medix over 20 baselines: it diminished the average FPR95 by 40.98% relative to formidable baselines such as KNN+, outstripped state-of-the-art methods such as WOODS and DRL, and attained an outlier extraction error rate as low as 12.5%.

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A Additional Studies

A.1 Element-wise Median vs Geometric Median

A further pertinent question that presents itself is why we should have elected the element-wise median in preference to the geometric median (Acharya et al., 2024) for the purpose of outlier detection in Medix. To resolve this question, we conducted a preliminary experiment employing CIFAR-10 as the InD dataset and SVHN as the OOD dataset. Specifically, $\mathcal{S}_{\text{wild}}$ comprises 5k samples drawn from CIFAR-10, so constituted as to be disjoint from the training set used to train the model $\phi_{\mathcal{S}_{\text{in}}}$ —the model from which we compute $\bar{\nabla}_{\text{in}}$. Incrementally introducing SVHN OOD samples into $\mathcal{S}_{\text{wild}}$, we track both the evolution of the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median of the gradients of the wild dataset and the proportion of OOD samples removed. The results, adduced in Figure 3, demonstrate that, for a fixed number of OOD samples in the wild dataset, the element-wise median designated as outliers a markedly higher proportion of OOD samples than did the geometric median. This finding indicates that the element-wise median is the more sensitive to distributional shifts and hence constitutes the more effective and reliable choice for the filtering employed in our method.

Table 3: The influence exerted by the hyperparameters ϵ and k upon OOD detection.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.2 Hyperparameter Selection and Sensitivity Analysis

We undertook an ablation study in order to gauge the sensitivity of Medix to the hyperparameters ϵ and k , employing for this purpose CIFAR-100 as the InD dataset and SVHN as the OOD dataset. As Table 3 attests, Medix exhibits pronounced robustness to variation in the values of ϵ and k . In particular, it attains a remarkably low FPR95 of 0.16 under the setting $\epsilon = 0.005$ and $k = 20000$; and even as the values of ϵ and k are perturbed—by, for instance, reducing ϵ to 0.0005 or halving k to 10000—the performance remains competitive (FPR95 of 0.20 and 0.68, respectively), thereby manifesting a graceful degradation rather than a precipitous collapse. These results indicate that, although judicious hyperparameter selection is not without importance, Medix preserves its effectiveness across a spectrum of hyperparameter choices and continues to surpass the baselines throughout. Such insensitivity to exact hyperparameter tuning renders Medix a dependable choice for real-world deployments, wherein exhaustive tuning may well prove infeasible.

A.3 Comprehensive comparison with recent competitive methods

We have broadened our comparisons so as to embrace competitive baselines that avail themselves of diverse contrastive learning objectives, among them SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023). Although these methods rest upon contrastive learning objectives, from which our own method abstains, we have included them in the interest of a comprehensive comparison; and to render the evaluation yet more thorough, we have further

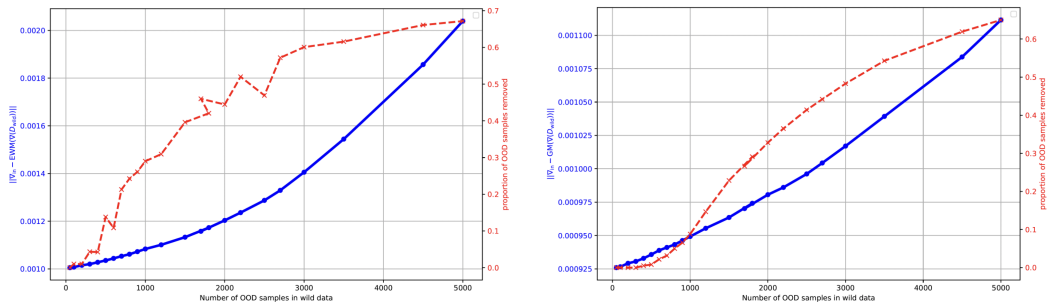


Figure 3: A comparison between the element-wise median (EWM) and the geometric median (GM).

Table 4: Comparison of OOD detection performance between Medix and recent competitive baselines with CIFAR-100 as the InD data. Performance is averaged over five runs, and the best results in each column are rendered in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

incorporated such recent methods as CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c). For this experiment we employ CIFAR-100 as the InD dataset and train the ResNet-34 model¹ in accordance with the setup of Ming et al. (2023); Zhang et al. (2024). The model is trained by stochastic gradient descent with a momentum of 0.9 and a weight decay of 10^{-4} over 500 epochs, the initial learning rate being 0.5 under cosine scheduling and the batch size 512; we make use of the checkpoints furnished by Ming et al. (2023)². The results, as set forth in Table 4, demonstrate that Medix outstrips even these competitive baselines—including those trained under contrastive learning objectives—by a substantial margin, attaining an average FPR95 of 8.68% and an average AUROC of 97.76%. Beyond this, Medix affords a provably low error rate, thereby extending theoretical guarantees that the majority of alternative approaches do not.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next inquire whether Medix is capable of accommodating large-scale wild OOD data within the unseen OOD setting. This setting proves the more exacting for two principal reasons: first, it draws upon a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), thereby amplifying both the complexity of the data and the attendant computational demand; and second, the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is advisedly chosen to differ distributionally from the wild OOD data marshalled during training—that is, $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ —a configuration that more faithfully mirrors realistic and challenging deployment scenarios. We adopted CIFAR-100 as the labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data, and the SVHN dataset as the test OOD data. This configuration introduces a heightened degree of complexity, precisely because the large-scale wild OOD data (300K Random Images) differ markedly from the OOD test data (SVHN). In order to appraise the performance of our approach, we set it against baselines that likewise draw upon wild data during training. The results reported in Table 5 throw into relief the superior performance of our method, Medix, relative to the baselines—evidenced by a substantially lower FPR95 (41.29 ± 1.2) together with a higher AUROC (87.25 ± 0.6)—and thereby attest to its efficacy in discriminating between InD and OOD data.

Table 5: A comparison of OOD detection performance on large-scale, unseen OOD data.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

²<https://github.com>

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

A further question of consequence is whether the quality of the pseudo-labels $\hat{y}_{\tilde{x}_i}$ —particularly as measured by softmax confidence—bears upon the performance of OOD filtering, and whether a simple pre-filtering step that excises low-confidence pseudo-labels might augment the robustness of the model. To resolve this question, we conducted an experiment gauging the effect of filtering low-confidence pseudo-labels upon OOD filtering, employing for this purpose CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. Specifically, we computed the softmax probabilities of each pseudo-labeled sample and discarded those whose predictions fell below a confidence threshold of 0.6; upon completion of this filtering, we found that 15.98% of the samples had been removed from the training set.

The results disclosed a virtually imperceptible difference in performance between the method with filtering and that without: both variants yielded an FPR95 of 0.01%, though with filtering the AUROC rose marginally from 99.98% to 99.99%. It follows that the removal of low-confidence pseudo-labels exerts no material influence upon the robustness of the model or upon its capacity to detect OOD samples—a finding that testifies to the resilience of our method in the face of noisy or low-confidence labels and intimates that further filtering steps are unlikely to yield any meaningful improvement.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We now address the question of whether the filtering phase of Medix is computationally feasible and memory-efficient when applied to large datasets. In response, we conducted profiling experiments on an NVIDIA A100-SXM4-80GB GPU using approximately 15,000 unlabeled samples for each InD–OOD pair, employing CIFAR-10 and CIFAR-100 as the InD datasets and LSUN-Resize as the OOD dataset. As the results in Table 6 make evident, GPU memory usage remains modest and the filtering phase concludes within a tractable timeframe, even when thousands of unlabeled samples are involved. This demonstrates that the Medix filtering process is computationally feasible and imposes no excessive memory overhead, even upon large datasets.

Table 6: Profiling results for the Medix filtering procedure on an NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Our work departs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in several cardinal respects. By way of illustration, in OpenML Saito et al. (2021) the central problem consists in coping with outliers in semi-supervised learning (SSL) while training a standard classifier, whereas in our setting the paramount challenge is to detect OOD samples from unlabeled wild data and to train a dedicated OOD detector; put otherwise, whereas the SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) aspire to improve classification in spite of outliers, our approach enhances OOD detection within an open-world setting in which labeled OOD data are unavailable.

In the second place, these SSL methods seek to train a classifier that is robust to the OOD samples arising in SSL, treating outliers as so much noise, whereas in our setting the objective is to detect OOD samples explicitly and to train an OOD detector. Stated differently, the SSL methods are preoccupied with suppressing OOD samples during training so as to improve SSL performance, whereas Medix actively detects and exploits these very OOD samples in order to construct a more effective and reliable OOD detection system.

Lastly, the two approaches diverge in their underlying techniques: OpenMatch is founded upon consistency regularization, the animating idea of which is to enforce consistency across disparate stochastic transformations of the same input—an idea mathematically instantiated through soft constraints on the classifier’s outputs that encourage smooth decision boundaries. Medix, by contrast, relies upon median-based filtering for outlier detection, marshalling the statistical robustness properties

791 of the median and its insensitivity to extreme values in order to identify prospective OOD samples;
792 its theoretical analysis, moreover, derives error bounds predicated upon the contamination effect
793 (the proportion of OOD samples) and the concentration effect (the sub-Gaussian behaviour of InD
794 gradients), thereby ensuring that the OOD detection error remains low.

795 **B Broader Impacts and Limitations**

796 As machine learning continues its advance, the confrontation of the challenges inherent in OOD
797 detection has become indispensable to the assurance of robust and reliable model performance within
798 real-world applications. We introduce a median-centric framework for OOD detection that enhances
799 both the handling of OOD inputs and the safety of the model. The ramifications of our research
800 extend well beyond the purely theoretical, admitting of practical application in domains as varied
801 as healthcare, autonomous systems, and finance; by ameliorating OOD detection, we address a
802 pivotal challenge attending model deployment and thereby foster a greater measure of trust in, and
803 adoption of, machine learning technologies. Future work might explore its integration with generative
804 models for the synthesis of outliers, or else its adaptation to dynamic environments in which the OOD
805 distributions evolve over time.

806 A prospective limitation of Medix resides in its reliance upon a mixture of unlabeled InD and OOD
807 data, which in practice may prove noisy, corrupted, or inconsistently sampled. This assumption is,
808 however, in no way peculiar to our method; it rather reflects the uncertainty intrinsic to real-world
809 deployment environments and constitutes a customary point of departure within robust learning
810 theory. So as to render our assumptions explicit and verifiable, we formalize them rigorously in
811 the main theorems and, in Remark 4.3, furnish a precise geometric condition that vindicates the
812 separation requisite for reliable filtering. We furthermore extend our guarantees to looser settings in
813 Theorem C.3, dispensing with the need for sub-Gaussian tails and thereby demonstrating the resilience
814 of our framework under weaker distributional assumptions. These endeavours serve to underscore
815 that, although Medix does indeed invoke simplifying assumptions, those assumptions are at once
816 interpretable and relaxable, and hence constitute a principled foundation for future improvements in
817 OOD detection under realistic noise conditions.

818 C Proof of Theorem

819 In order to promote clarity and consistency throughout the theoretical and empirical sections alike,
 820 we consolidate the principal notation employed in this paper within Table 7, which encompasses
 821 definitions of the gradient-related quantities, the sample sizes, and the filtering terms, together with
 822 the parameters that figure in our concentration bounds. Unless stated to the contrary, we reserve
 823 boldface for vectors, denote the input space by $\mathcal{X}$, and assume that all gradients are computed with
 824 respect to a fixed pretrained model f_ϕ .

Table 7: A summary of the notation. Bold symbols are reserved for vectors, and $\mathcal{X}$ denotes the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors

825 C.1 Bound on the Inlier Misclassification Rate

826 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 827 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 828 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 829 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

830 *Proof.* We articulate our argument in two principal steps. In **Step 1** we establish a high-probability
 831 concentration bound for the gradients of the InD samples about their mean $\bar{\mathbf{V}}_{\text{in}}$, designating as a *good*
 832 *inlier* any InD sample whose gradient falls within this high-probability region. In **Step 2** we invoke
 833 a “swapping” argument upon the EWM objective so as to demonstrate that the removal of more
 834 than a $(2\eta + \frac{\pi}{2}(1 - \pi))$ fraction of the *good inliers*—where η denotes the fraction of InD samples

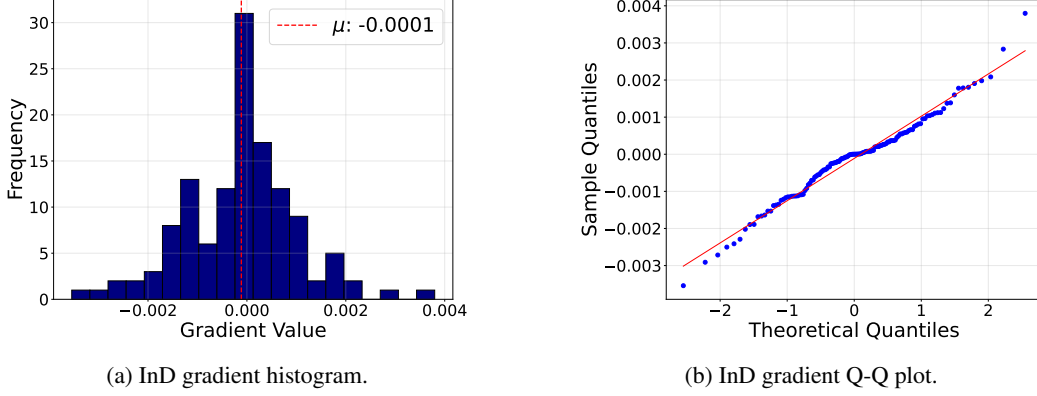


Figure 4: An illustration of the sub-Gaussian behaviour exhibited coordinate-wise by the gradients of InD samples. (left) A histogram of gradient values (CIFAR-100 InD data), which manifests concentration about the mean together with light tails, consistent with sub-Gaussianity. (right) A Q-Q plot juxtaposing the empirical quantiles of the InD gradients against a theoretical Gaussian distribution, which confirms their alignment with sub-Gaussianity.

whose gradients lie outside the high-probability region (that is, the non-good InD samples)—would
 controvert optimality. Amalgamating these two steps, we thereupon derive the stated upper bound on
 the inlier misclassification rate ERR_{in} .

Step 1: Concentration of InD Gradients. Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points within the set $\mathcal{S}_{\text{wild}}$. We deem an InD
 point $\mathbf{x}$ to be *bad* whenever

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

Our objective is to establish that, with high probability, only a small fraction of these m_{in} inliers can
 be bad.

From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define
 a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

Here $\mathbf{x}$ denotes a generic InD sample, and we assume the samples to be i.i.d. across the m_{in} inliers.

Let X denote the random variable counting the number of bad inliers among the m_{in} samples. For
 each inlier $\mathbf{x}_i$, we define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

854 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

855 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

856 Observing that X constitutes a sum of m_{in} Bernoulli(p) random variables, each confined to the
857 interval between 0 and 1, we may invoke Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

858 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

859 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

860 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

861 Dividing both sides by m_{in} thereupon yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

862 Let us denote this fraction by η . Setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, we thereby conclude that *with*
863 *probability at least $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of the inliers*
864 *possess gradients lying beyond the radius $\epsilon\sqrt{d}$ about $\bar{\nabla}_{\text{in}}$. These are precisely the bad inliers, whereas*
865 *the remainder are termed good inliers.*

866 **Step 2: Contradiction via a Swapping Argument** We demonstrate that, should $\mathcal{S}^*$ exclude an
 867 inordinate number of InD points—specifically, whenever

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

868 —then one may construct a new set $\widehat{\mathcal{S}}$ whose EWM lies strictly nearer to $\bar{\nabla}_{\text{in}}$ than does that of $\mathcal{S}^*$,
 869 thereby contravening the optimality of $\mathcal{S}^*$.

870 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 871 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

872 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

873 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

874 Define the swap size as

$$m'_\varepsilon := \min \{ m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (26)$$

875 This quantity enumerates how many good inliers (those situated within B_ε) are at present excluded
 876 from $\mathcal{S}^*$ and thus available to be swapped in.

877 We construct a new set $\widehat{\mathcal{S}}$ by excising from $\mathcal{S}^*$ the m'_ε points that lie outside B_ε (OOD points or
 878 bad inliers) and inserting in their stead m'_ε good inliers drawn from B_ε —an operation that preserves
 879 cardinality:

$$|\widehat{\mathcal{S}}| = |\mathcal{S}^*|.$$

880 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

881 which gauges the fraction of good inliers excluded from $\mathcal{S}^*$. Upon rearrangement, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

882 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

883 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

884 We know that at most an η fraction of the inliers reside outside B_ε , while the fraction of OOD points
 885 among all samples is π ; accordingly, the total number of samples in the dataset that lie outside B_ε
 886 satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

887 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)}\right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi}\right) m_{\text{in}} \geq \frac{m_{\varepsilon}}{2}. \quad (32)$$

888 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_{\varepsilon}}{2}, \quad (33)$$

889 So $m'_{\varepsilon} = m_{\varepsilon}/2 + \eta m_{\text{in}}$, which implies

$$m_{\varepsilon} - m'_{\varepsilon} = \frac{m_{\varepsilon}}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_{\varepsilon} = m_{\text{in}}^* + \frac{m_{\varepsilon}}{2} + \eta m_{\text{in}}. \quad (35)$$

890 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_{\varepsilon} > m_{\varepsilon} - m'_{\varepsilon}, \quad (36)$$

891 so that the points reposing within B_{ε} constitute a strict majority in $\hat{S}$.

892 Fix any coordinate $j \in \{1, \dots, d\}$, and let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ denote the j -th coordinate values
893 of all gradients in $\mathcal{S}^*$, with $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ their counterpart in $\hat{S}$, and set $\mu_j = \bar{\nabla}_{\text{in},j}$.

894 Since each inserted gradient lies within the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$ while each removed gradient
895 lies outside I_j , it follows that, for some coordinate j^* , the number of values inside I_{j^*} ascends from
896 below $m/2$ to above $m/2$ by virtue of the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

897 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

898 As for the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unaltered or
899 augment it, so that the median distance from μ_j does not increase and may indeed diminish.

900 Thus, in a component-wise sense:

$$|\text{EWM}_j(G_{\hat{S}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

901 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{S}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

902 This contravenes the assumption that $\mathcal{S}^*$ constitutes a minimizer of the optimization problem, and
903 therewith the proof is concluded.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

904 □

905 C.2 Bound on the Outlier Retention Rate

906 We now establish a non-asymptotic upper bound on the fraction of OOD points that persist within
907 the subset returned by the MEDIX filter. This result mirrors the inlier misclassification bound
908 (Appendix C.1); yet, rather than relying upon inlier concentration, it capitalizes upon a separation
909 assumption obtaining between the OOD and InD gradient means.

910 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
 911 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

912 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

913 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

914 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

915 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

916 *Proof. Step 1: Concentration of OOD Gradients.*

917 By hypothesis, each coordinate of $g(x)$ is sub-Gaussian and the mean vector stands separated from
 918 $\bar{\nabla}_{\text{in}}$ by no less than $\Delta\sqrt{d}$. Accordingly, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

919 has probability at most $p_{\text{out}}(\varepsilon)$.

920 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

921 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

922 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

923 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

924 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

925 which holds with probability at least $1 - \delta$.

926 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

927 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_\varepsilon^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus more than half of the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients repose outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the very definition of the coordinate-wise median, this entails that, in every coordinate, the majority of the entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ fall outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, and hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Consider now a new subset $\widehat{\mathcal{S}}$, obtained by exchanging the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ for an equal number of InD points whose gradients reside within B_ε . Such inliers are guaranteed to exist, inasmuch as fewer than $m_{\text{out}}\eta_{\text{out}}$ of them are ambiguous while the total complement of InD points amounts to m_{in} .

Thereupon at least half the entries in every coordinate of $G_{\widehat{\mathcal{S}}}^{(j)}$ come to lie within the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, which implies:

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\widehat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contravenes the optimality of $\mathcal{S}_{\text{in}}^*$, whence the assumption must be false. Consequently, with probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla\ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla\ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

948 **Differences with the original bound.** The original bound, predicated as it is upon sub-Gaussianity,
 949 exhibits a decay term

$$2 \exp\left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

950 that diminishes exponentially in the dimension d and thereby renders the bound exceedingly tight for
 951 large d . The new bound, by contrast, which rests solely upon the assumption of finite fourth moments,
 952 displays a decay term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}$$

953 that decays merely as $O(1/d)$ and hence yields a looser bound for larger d . Whereas the original
 954 bound demands sub-Gaussian tails, the new one presupposes no more than finite fourth moments and
 955 so admits heavier tails; both bounds, however, share the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

956 which contracts as m_{in} grows, independently of the tail assumptions.

957 *Proof.* The proof adheres to the architecture of the original Theorem 4.1, comprising two principal
 958 steps: first, the establishment of a concentration bound on the fraction of “bad” InD samples (those
 959 lying far from ∇_{in}) without any recourse to sub-Gaussianity; and second, the deployment of a
 960 swapping argument to bound the inlier misclassification rate. The salient difference consists in
 961 the substitution of the sub-Gaussian tail bound by a moment-based bound derived from the finite
 962 fourth-moment assumption.

963 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

964 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 965 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon\sqrt{d}, \quad (52)$$

966 and “good” otherwise. Our aim is to bound the fraction of bad inliers among the m_{in} InD samples in
 967 S_{wild} .

968 Inasmuch as $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm by way of the sum of squared coordinates.
 969 We assume that the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each
 970 satisfying:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

971 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

972 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

973 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

974 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

975 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

976 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

977 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

978 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
979 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

980 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
981 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

982 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

983 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

984 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

985 This quantity η constitutes the upper bound on the fraction of InD samples whose gradients deviate
 986 from $\bar{\nabla}_{\text{in}}$ by more than $\epsilon\sqrt{d}$, a bound predicated solely upon finite fourth moments together with
 987 coordinate independence.

988 **Step 2: Contradiction via a Swapping Argument** We demonstrate that, should $\mathcal{S}^*$ exclude an
 989 inordinate number of InD points—specifically, whenever

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

990 —then one may construct a new set $\hat{\mathcal{S}}$ whose EWM lies strictly nearer to $\bar{\nabla}_{\text{in}}$ than does that of $\mathcal{S}^*$,
 991 thereby contravening the optimality of $\mathcal{S}^*$.

992 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 993 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

994 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

995 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

996 Define the swap size as

$$m'_\epsilon := \min \{ m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (71)$$

997 This quantity enumerates how many good inliers (those situated within B_ϵ) are at present excluded
 998 from $\mathcal{S}^*$ and thus available to be swapped in.

999 We construct a new set $\hat{\mathcal{S}}$ by excising from $\mathcal{S}^*$ the m'_ϵ points that lie outside B_ϵ (OOD points or
 1000 bad inliers) and inserting in their stead m'_ϵ good inliers drawn from B_ϵ —an operation that preserves
 1001 cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

1002 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

1003 which gauges the fraction of good inliers excluded from $\mathcal{S}^*$. Upon rearrangement, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

1004 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

1005 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

1006 We know that at most an η fraction of the inliers reside outside B_ε , while the fraction of OOD points
 1007 among all samples is π ; accordingly, the total number of samples in the dataset that lie outside B_ε
 1008 satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

1009 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

1010 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

1011 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

1012 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

1013 so that the points reposing within B_ε constitute a strict majority in $\widehat{\mathcal{S}}$.

1014 Fix any coordinate $j \in \{1, \dots, d\}$, and let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ denote the j -th coordinate values
 1015 of all gradients in $\mathcal{S}^*$, with $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ their counterpart in $\widehat{\mathcal{S}}$, and set $\mu_j = \bar{\nabla}_{\text{in},j}$.

1016 Since each inserted gradient lies within the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$ while each removed gradient
 1017 lies outside I_j , it follows that, for some coordinate j^* , the number of values inside I_{j^*} ascends from
 1018 below $m/2$ to above $m/2$ by virtue of the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

1019 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

1020 As for the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unaltered or
 1021 augment it, so that the median distance from μ_j does not increase and may indeed diminish.

1022 Thus, in a component-wise sense:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

1023 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

1024 This contravenes the assumption that $\mathcal{S}^*$ constitutes a minimizer of the optimization problem, and
 1025 therewith the proof is concluded.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

1026 □